Challenges in 2D Hybrid Non-Quasi-Neutral Hall Thruster Simulation

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***Abstract*—Hall thrusters (HTs) are the most widely used electric propulsion systems in development today. However, accurate modeling remains challenging due to problem stiffness, complex multiphysics interactions, and poorly understood particle-level phenomena inherent in HT plasmas. For the past two decades, the standard has been to prioritize computational efficiency by assuming a quasi-1D electron fluid and quasi-neutral electric field. In this work, we review the assumptions behind this standard model, examine the computational trade-offs of reintroducing more complete physics, and present early qualitative successes from a 2D hybrid-PIC simulation incorporating full Euler fluid dynamics, a Poisson solver, and DSMC based fluid-to-particle collisions.**

***Index Terms*—Hall thrusters, computational fluid dynamics, hybrid model, PIC, Poisson.**

I. Introduction

Hall thrusters (HTs) are the most widely researched and flight-proven form of electric propulsion today, offering a powerful balance of efficiency, versatility, and performance [1], [2].

Operationally, Hall thrusters generate thrust by rapidly accelerating ions through an axial electric field. Electrons are injected at an external cathode and become trapped in a rapidly gyrating azimuthal cloud due to the combined effects of the applied voltage and magnetic field [3]. A high-Z noble gas, such as xenon, is injected at the anode and ionized through impact collisions with these high-energy electrons. The resulting ions are then accelerated downstream at velocities exceeding 30 km/s, producing a high-velocity plume with specific impulses over 3000s and power levels between 10 and 100kW [4]. These characteristics place Hall thrusters in the critical middle ground between high-thrust and high-efficiency propulsion, cementing their role in deep space exploration and long-duration space transport [5].

HT simulation efforts began in the 1990s using fully kinetic models and Poisson solvers to resolve particle-scale physics, but these were limited to the near-plume region due to computational cost [6]. By the early 2000s, hybrid-PIC approaches emerged, combining fluid and kinetic models with simplifying assumptions such as quasi-neutrality, quasi-1D electron fluids, and inertialess momentum, which enabled higher resolution simulations [7], [8]. More recently, fluid-only solvers have been developed to improve speed and extend coverage into the far plume [9]. Though well-suited for industrial use, none of these methods have successfully modeled all aspects of Hall thruster particle-scale physics [3].

A major challenge is that plasma simulations are inherently unstable due to extreme disparities in spatial and mass scales—i.e., the problem is fundamentally stiff [10]. Compounding this, bulk Hall thruster performance is strongly affected by anomalous electron transport, a phenomenon not predicted by classical models and thought to arise from particle-level irregularities that remain difficult to simulate [3], [11].

With modern computational resources, however, many of the simplifying assumptions used in earlier simulations can now be revisited to design more precise models. Recent work has reintroduced high fidelity techniques such as full Poisson solves [12], [13], non-quasi-neutral electron fluid models [14], and even fully kinetic treatments [15] while achieving runtimes significantly shorter than those required just a decade ago.

In this work, we explore the reintroduction of several such precise methods in the hybrid-PIC `AXCFD` code. Our code implements a full Poisson solver for the electric field, non-approximated 2D Euler equations for the electron fluid, and an experimental fluid-to-particle Direct Simulation Monte Carlo (DSMC) collision mechanism. We review the challenges of

applying these methods, their impact on performance and efficiency, and present early results from a hybrid-PIC framework designed to model HT physics more completely and accessibly than conventional hybrid quasi-neutral approaches.

II. Particle Methods in AXCFD

AXCFD is a hybrid-PIC HT simulation framework built on compressible fluid solvers. In this section, we discuss the particle and multi-fluid coupling methods.

A. Particle-In-Cell Integration

Multi-species simulation is essential for capturing the distinct dynamics and scale separations present in HT plasmas. To this end, AXCFD separately models the heavy ion and neutral species using an operator-split Particle-In-Cell (PIC) method [16]. In addition, kinetic models such as PIC are capable of resolving non-Maxwellian particle behavior in the plasma sheaths, reducing critical instability growth in HTs [12].

In the PIC framework, similar particles in phase space are grouped as *macroparticles*, each carrying a statistical weight. This weight is assigned at initialization or injection according to:

$$q_{f,\alpha} = \frac{n_{p,\alpha} A_c}{N_{\text{ppc}}} \tag{1}$$

where, $n_{p,\alpha}$ is the target number density of species α, A_c is the area of a computational cell, and N_{ppc} is the desired number of macroparticles per cell. Recent axial-azimuthal simulations have found an N_{ppc} of 100–200 ideal for statistical convergence [17].

We now shift our attention to particle dynamics. Because xenon ions are effectively unmagnetized in HT plasmas [3], their trajectories are primarily governed by the Coulomb force given by Newton's second law:

$$m_\alpha \frac{d\vec{v}}{dt} = q_\alpha \mathbf{E} \tag{2}$$

where m_α is the mass of the particle, q_α is the particle charge, and $\mathbf{E}$ is the interpolated electric field at the particle position. Particle motion is integrated using a forward Euler scheme at each timestep.

B. Physical and Chemical Collisions

To model collisions between species, we employ a Direct Simulation Monte Carlo (DSMC) method, where collision probabilities depend on the energy of incident particles [15]. AXCFD accounts for e-n (electron-neutral) scattering, e-n excitation, e-n ionization, and i-n (ion-neutral) scattering.

For each particle pair, we check whether the incident electron or ion exceeds the kinetic energy threshold for the collision process, E_β, as defined in the LXCat plasma collisions database [18], and compute the collision frequency as:

$$\nu_\beta = n_t \sigma_\beta(E_p) v_p \tag{3}$$

where n_t is the local neutral density, $\sigma_\beta(E_p)$ is the energy-dependent cross-section for collision type β, E_p is the incident particle energy, and v_p is the relative velocity between particles. The probability of collision β is thus $P_\beta = 1 - \exp(-\nu_\beta \Delta t)$.

If $R < P_\beta$ for a randomly drawn number $R \in [0, 1]$, the collision is successful. Appropriate energy loss, momentum scattering, and continuity are then applied to both particles. Otherwise, the Monte Carlo algorithm attempts the next applicable collision type.

III. Confronting Fluid Simplifications

We now review the most common assumptions made in state-of-the-art HT simulations regarding the electron fluid and field models, and examine the practical implications of reintroducing their full formulations in AXCFD.

A. Quasi-neutral and Quasi-1D Electron Models

A common simplification in popular HT simulations like HPHall-2 [8] and Hall2De [9] is the assumption of bulk quasi-neutrality, i.e., $n_e = n_i$. This reduces both the continuity and momentum equations to trivial time-independent algebraic expressions, which form the basis of the so-called *drift-diffusion* model.

Most drift-diffusion formulations close the system using magnetic-field-aligned expressions for the electric field [9]. By simplifying both momentum and electric field into algebraic expressions, this approach uniquely avoids the significant numerical stiffness caused by their disparate mass scales. However, this approach causes field-aligned cells near boundaries to often have highly skewed or disproportionate geometries, which can degrade accuracy [3]. Moreover, algebraic sheath models are typically required to capture the non-quasi-neutral charge separation that occurs in sheath regions [15]. To avoid this complexity, some codes neglect cross-field dynamics entirely and only resolve axial transport under the *quasi-1D* assumption [7].

AXCFD chooses to retain full 2D resolution of the electron fluid by solving the unsimplified compressible Euler equations augmented with electromagnetic source terms based on the derivation in [12]:

$$\frac{\partial \mathbf{U}_e}{\partial t} + \nabla \cdot \mathbf{F}_e = \mathbf{S}_e \tag{4}$$

$$\mathbf{S}_e^{\rho u} = \frac{q_e}{m_e} \rho_e (\mathbf{E} + \mathbf{V}_e \times \mathbf{B}) \tag{5}$$

$$\mathbf{S}_e^{E} = \frac{q_e}{m_e} \rho_e \mathbf{E} \cdot \mathbf{V}_e \tag{6}$$

where $\mathbf{U}_e$ are the Eulerian conserved quantities, $\mathbf{F}_e$ is the flux vector, ρ_e is the electron density field, $\mathbf{V}_e$ the 2.5D electron velocity field, and $\mathbf{B}$ the global magnetic field. Coupled with the ion PIC model, this formulation eliminates the need for artificial sheath models or explicit treatment of the pre-sheath transition near boundaries. The inclusion of electron inertia also facilitates the emergence of nonlinear effects such

as plasma oscillations and anisotropic transport, which are hypothesized to underlie anomalous electron transport [15].

B. Quasi-neutral Electric Field

Per the definition of a *quasi-neutral* plasma model, all quasi-neutral simulations approximate $\mathbf{E} = 0$. This condition holds in HT plasmas at scales larger than the Debye length, and is generally sufficient for industrial applications [3], [12]. However, this simplification fails to capture charge separation effects at the sheaths. As a result, many hybrid codes must again rely on approximate models to resolve sheath behavior [19], limiting their ability to self-consistently capture important anomalies at the channel walls.

In contrast, our model solves the electric field from first principles to better resolve these surface anomalies. The field is directly related to species charge density via the Poisson equation for electrostatics:

$$\nabla^2 \phi = \frac{\rho}{\epsilon_0} \tag{7}$$

where ρ is the species charge density, ϵ_0 is the vacuum permittivity, and ϕ is the electric potential. We solve this linear system using the Unsymmetric Multi-Frontal Method and compute the electric field as $\mathbf{E} = -\nabla\phi$.

One practical reason Poisson solves are typically avoided is that they can account for over 80% of total CPU usage [12]. We mitigate this by using a low-resolution mesh and limiting the number of particles contributing to charge deposition.

C. Collision Operators

Most hybrid HT simulations do not resolve collisions using the particle-based DSMC method, since electrons are modeled as a fluid [19]. Instead, they rely on analytical collision operators derived from classical and anomalous collision frequency models. The refinement and analysis of these operators remains an active area of research, particularly due to their importance in modeling anomalous transport [3], [20].

In contrast, to support the higher-fidelity DSMC collisions, `AXCFD` introduces an experimental fluid-to-particle seeding mechanism that converts the electron fluid into particles to allow for full DSMC calculations at every timestep. As is typical in plasma physics, we seed particles assuming a Maxwellian distribution [12]. We then compute the space-averaged sum of all electron collisional cross-sections and apply the result as a collision damping term to the electron fluid in the following timestep.

IV. Example: 2D HT Channel Simulation

To validate `AXCFD` and explore its ability to capture complex plasma behavior, we revisit the popular HT channel problem from the late 1990s [21]. This setup proves ideal for testing an early version of our solver, as it abstracts away logistical complexities such as cathode injection, thrust measurement, and physical boundaries. Rather, the channel problem focuses on the essential physics of the thruster: ionization, ion acceleration, electron curvature, electric field–charge density interaction, and particle collisions.

Despite this simplification, the underlying physics remains immensely complex. As a result, early simulations required normalization, heuristic adjustments, and careful tuning for stability. We therefore present the problem setup and early results in empirically scaled units.

A. Problem setup

We define a 2D simulation domain representing an axial-radial cross section of the HT channel, spanning the full axial length L. The grid for field-based quantities is a 50×20 Cartesian mesh. Electron variables are bounded by Neumann conditions on the anode and both walls. A dampened potential gradient is applied via a right-hand Dirichlet boundary set to 3V, with a zero-gradient Neumann boundary at the anode.

Electrons are initialized as a static fluid with normalized density, pressure, temperature, and charge. Neutrals are initialized with constant density ($99\rho_e$) and a target of 10 particles-per-cell (N_{PPC}). Inflow of neutrals is maintained at the anode at each timestep to preserve the PPC target. For numerical stability, heavy species are adjusted with a mass ratio $m_\alpha/m_e = 100$ and a temperature ratio $T_\alpha/T_e = 10$. A static magnetic field is imposed with a Gaussian-like radial profile, with $\mathbf{B}_{\text{max}} = 0.02$T.

The electric field is computed after each Poisson solve using second-order central differences applied to the electrostatic potential, and the resulting electric field spans the full channel.

To solve the electron-Euler equations, we use a Finite-Volume Method (FVM) with Rusanov fluxes, which offer favorable numerical dissipation. Time integration is performed using the first-order Runge-Kutta method.

The simulation was run for 175 timesteps using a CFL condition of 0.1 and a timestep $\Delta t = 900\mu s$.

B. Results

We compare our simulation results with experimental measurements for the Stanford Hall Thruster (SHT) [13]. Experimental data points have been vertically scaled to match the simulation's normalization and horizontally aligned with the magnetic field profile.

Figure 1 shows the axial variation of n_e, T_e, and ρ_e with the induced magnetic field highlighted in blue. The simulation demonstrates good qualitative agreement with experimental measurements for $z < 0.6$. The maximum in electron density at $z = 0.4$ indicates a high ionization rate correctly beginning at the onset of our applied magnetic field, which spans $0.4 < z < 0.8$. The electron temperature also correctly rises through the ionization region up to $z = 0.5$, then gradually declines post-ionization. Similar trends are observed in the electron-neutral collision frequency, though with notable numerical noise due to our conservative particles-per-cell (N_{ppc}) setting.

Beyond $z = 0.6$, the simulation begins to overestimate density and energy. Interestingly, this is consistent with trends observed in the axial-azimuthal hybrid simulation by [13], who also omitted an explicit cathode model. Both simulations seem to share constraints tied to the absence of realistic cathode physics. We believe this behavior stems from an

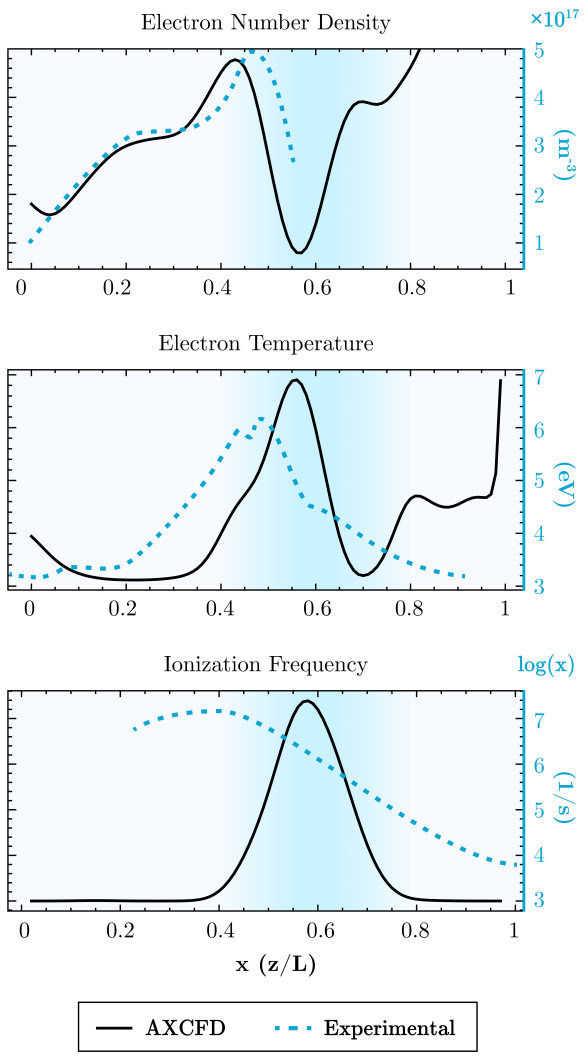


Fig. 1. Variation of electron number density, electron temperature, and ionization frequency over normalized axial distance. Experimental results from SHT data in [13]. The imposed magnetic field region is highlighted in blue.

overly restrictive cathode inflow condition, which forces high densities and energy gradients near the right boundary. A likely remedy, short of modeling the cathode directly, would be to extend the domain to buffer the core region from boundary effects. While we are exploring this solution, limitations in computational resources have so far prevented testing on larger domains.

Figure 2 shows the full 2D domain, where we observe an expected dip for electron density and a peak for the azimuthal (z-direction) velocity in the ionization cloud region.

A Gaussian filter with σ = 1.5 is applied to ion and neutral densities. Some noise remains in both fields as expected, but macroscopic trends are consistent with ion density peaking in the $0.4 < z < 0.8$ range and neutral density tapering from a maximum at the anode inflow.

The electric potential also correctly reflects the applied cathode voltage.

C. Electron-ion DSMC Results

We observe promising preliminary results from our experimental DSMC electron-conversion method. The electron collision cross-section field exhibits the expected structure: high-σ regions appear outside the ionization zone where frequent elastic scattering dominates, while a lower-σ region forms in the center, consistent with the prevalence of low-σ ionization and excitation events. This spatial pattern aligns qualitatively with LXCat cross-section data, supporting the physical accuracy of our collision modeling.

However, the interpolation of σ values back to electrons may require further refinement, as the electron fluid fails to remain confined as predicted. Rather than settling near the mid-channel and being repelled from the anode, electrons complete the circuit more freely, suggesting that the confining effect of collisions is underrepresented. Further empirical adjustment is necessary to refine and validate this capability.

D. Scaling and stability challenges

A central challenge in conducting these preliminary simulations was maintaining numerical stability. Plasma systems are notoriously stiff due to their extreme scale disparities (e.g., $m_e : \mathbf{E} \sim 10^{33}$), making normalization essential for tractable precision [10].

However, full self-consistent normalization was infeasible, as scaling one quantity required compensatory changes in others that preserved stiffness. Thus, to avoid needing to switch to costly quadruple-precision formats, we introduced a limited set of non-self-consistent scaling multipliers, heuristically tuned for stability.

For example, we reduced the electric field magnitude by $\sim 10^{34}$, then compensated with a 10^2 multiplier in the Lorentz force update, a 10^{18} factor in collision cross-section evaluation, and a 10^{10} increase in z-momentum energy terms. This produced reasonable ionization rates, stable electron dynamics, and manageable value ranges.

Unfortunately, such a system is inherently qualitative. We continue iterating upon this approach as we attempt to balance physics, numerical convergence, and stability.

V. Conclusion

HT plasma simulation remains a challenging problem due to its complex physics, critical particle-level interactions, and inherent stiffness. In this work, we reviewed common simplifications used in hybrid models and introduced `AXCFD`: an HT simulation that reintroduces key electron dynamics, electric field behavior, and DSMC-based collisions into the modern hybrid simulation landscape. `AXCFD`'s bulk-quantity predictions showed strong qualitative agreement with experimental results and recent non-cathode models. Our experimental collision algorithm also demonstrated qualitative accuracy. Looking ahead, immediate improvements include optimizing `AXCFD` for higher N_{ppc} values and finer grid resolutions to fully leverage its predictive physics capabilities. Incorporating EM-based CFL conditions as described in [12] is also a high priority. Finally, we plan to refine our stiffness-mitigation

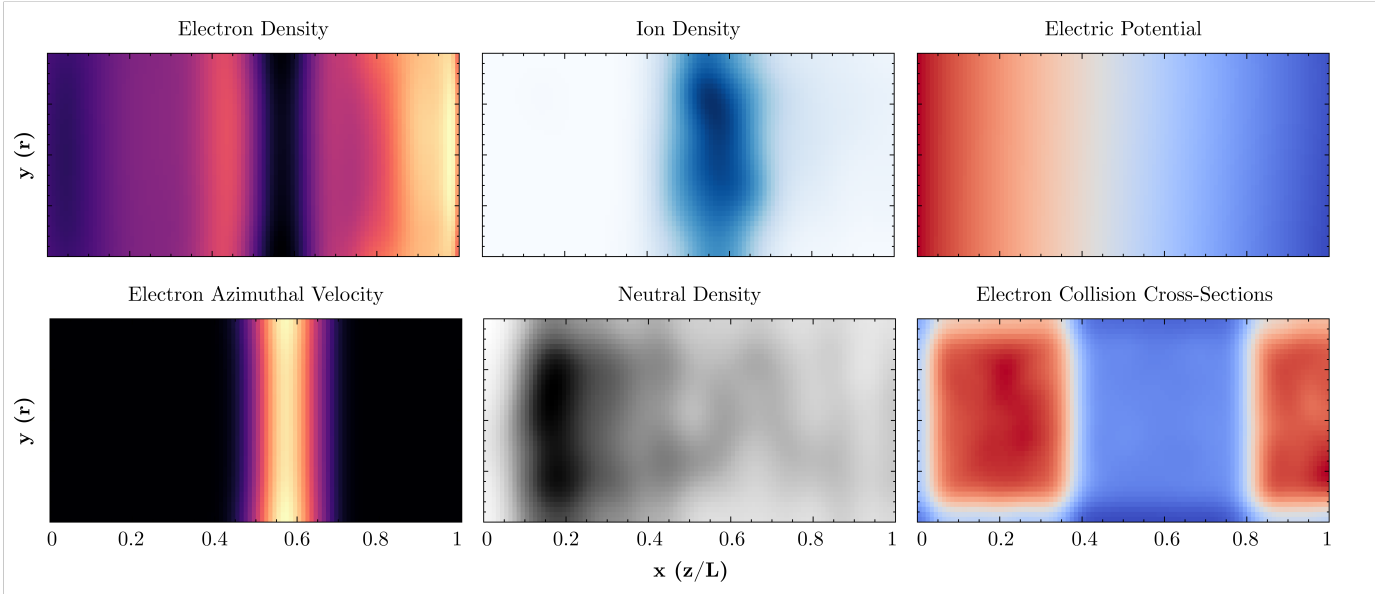


Fig. 2. Macroscopic electron density and electron θ-velocity (yellow high, purple low). Ion and neutral average densities (dark high, light low). DSMC-derived collision cross-sections and electric potential (red high, blue low). Data shown at timestep 175.

strategy through further tuning of heuristic multipliers. We do all this in pursuit of a fully physical model that surpasses quasi-neutral assumptions in this emerging class of complete-physics Hall Thruster simulations.

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